

Predicting Body Weight: The Impact of Height and Gender

A Statistical Analysis of the 2007 Youth Risk Behavior Survey (YRBS)

Research Question
Can a high school student's height and biological sex significantly predict their weight?

Variables
Dependent Variable (Y): Weight (kg)
Independent Variables (X): Height (meters) & Sex (Female/Male)

Methodology
Model: Multiple Linear Regression (OLS)
Sample Size (N): 13,062 students
Model Fit (R^2): 0.265 (This indicates that 26.5% of the variance in weight is explained by height and sex).

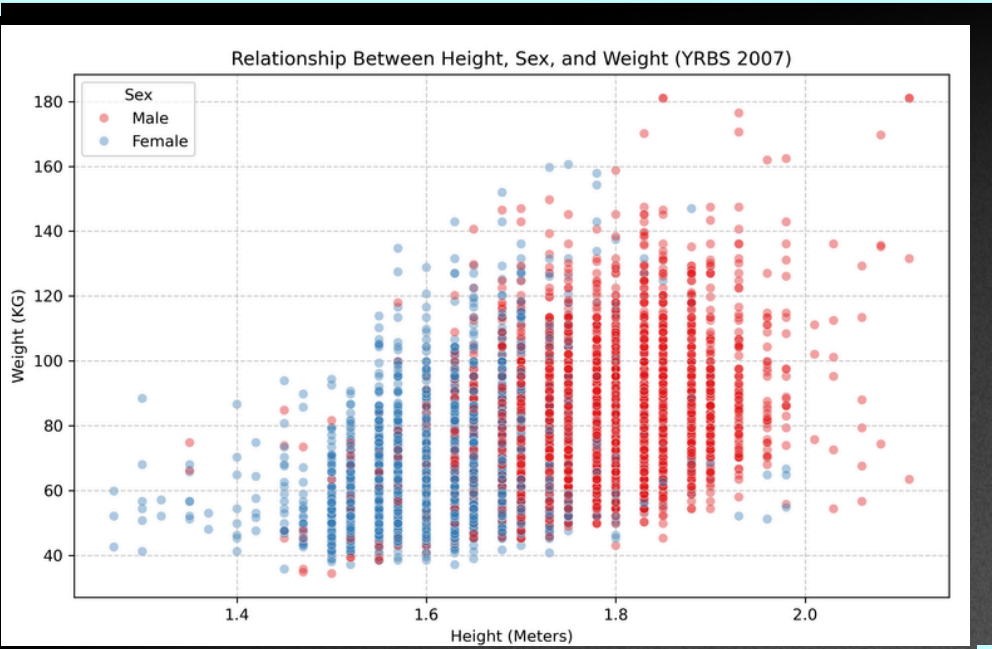


Figure 1: Scatter plot showing a positive correlation between height and weight for both genders.

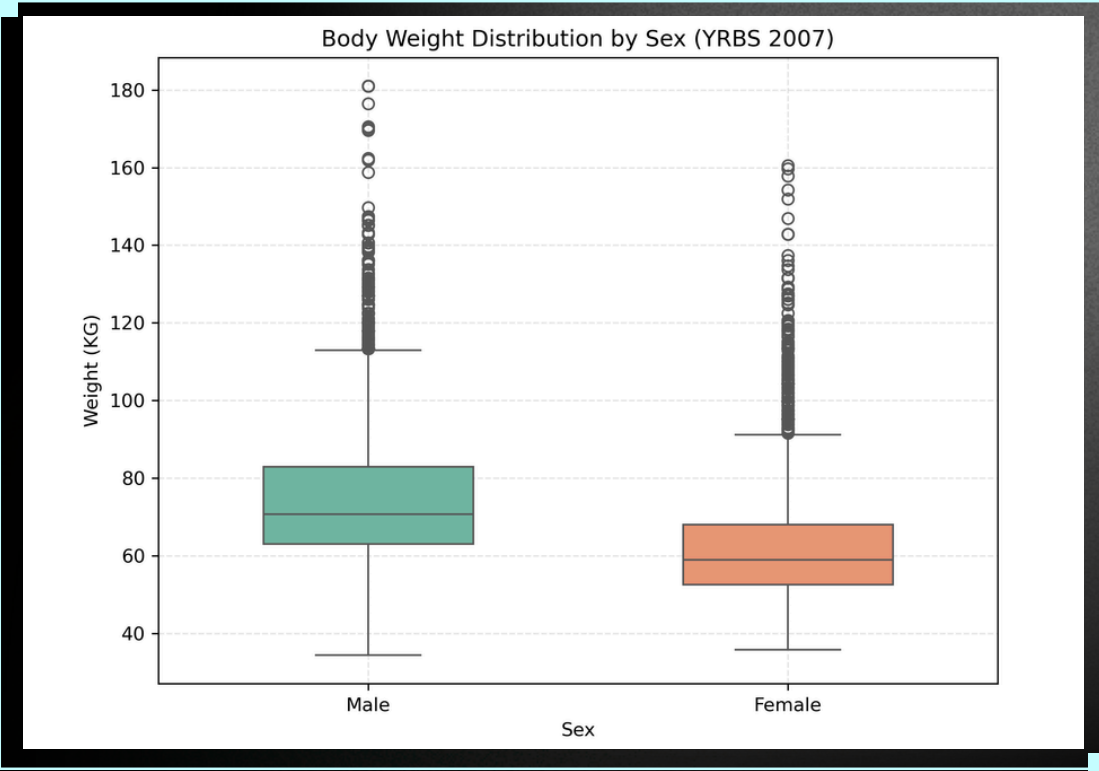


Figure 2: Box plot illustrating the higher median and wider variance in male students' body weight.

Key Findings:
Height is Highly Significant: There is a strong positive correlation between height and weight ($\beta = 76.57$, $p < 0.001$). For every 1-meter increase in height, a student's weight increases by an average of 76.57 kg.
Significant Gender Gap: Biological sex is also a significant predictor ($\beta = 3.65$, $p < 0.001$). Holding height constant, male students weigh on average 3.65 kg more than female students.
Overall Significance: Both variables successfully predict body weight, statistically confirming the physical trends observed in the visual data.

Conclusion:
The multiple regression model successfully demonstrates that a combination of height and biological sex is a robust predictor of body weight among high school students.